

Credit Card Default Prediction

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Abstract—This report presents a machine learning pipeline for credit card default prediction, framed as a binary classification problem evaluated with Macro F1 to equally weight both classes. Starting from a logistic regression baseline (Macro F1 = 0.628), this paper will report all the trials used in order to achieve some good results, in particular will be presented: preprocessing strategies, classifiers, Bayesian hyperparameter optimization, and cross-validated probability threshold calibration. The final submitted model is a tuned Random Forest trained on domain-specific engineered features (S3), achieving a cross-validated Macro F1 of 0.707 — a 12.6% relative improvement over the untuned baseline. A more complex configuration combining one-hot encoded categorical variables and a calibrated decision threshold ($t = 0.533$) reached 0.711 on cross-validation but generalised worse to the held-out evaluation set, so the simpler S3 configuration with $t = 0.5$ was retained as the final submission.

I. PROBLEM OVERVIEW

The task is to predict whether a credit card client defaults on their next monthly payment (*default.payment.next.month* $\in \{0, 1\}$) given 23 features: demographic attributes (sex, age, education, marriage status, credit limit), six months of repayment status (*PAY_0*–*PAY_6*), bill amounts (*BILL_AMT1*–*BILL_AMT6*), and payment amounts (*PAY_AMT1*–*PAY_AMT6*) [1]. The development set contains 24,000 samples split 80/20 into training and evaluation; no external data were permitted. The class distribution is 77.9% non-defaulters vs. 22.1% defaulters (3.52:1 ratio, Fig. 1), making Macro F1 the appropriate metric rather than accuracy.

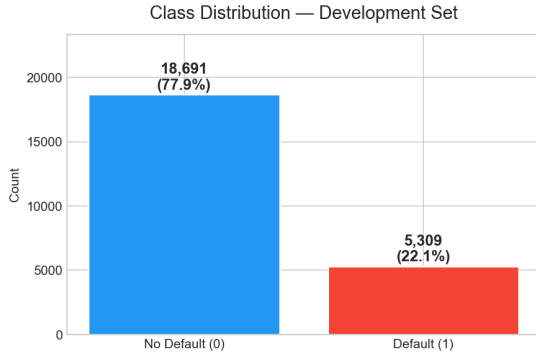


Fig. 1. Class distribution in the development set. Such an imbalance ratio motivates class-weighted training and threshold calibration.

A. Dataset Quality Issues

A preliminary inspection revealed several data quality issues requiring attention prior to modelling.

Undocumented categorical values. *EDUCATION* is defined for values 1 (graduate school), 2 (university), 3 (high school), and 4 (other), yet the dataset contains additional undocumented codes $\{0, 5, 6\}$ in roughly 0.5% of records. Similarly, the column *marriage* is documented for 1 (married), 2 (single), and 3 (other), but value 0 appears in a small fraction of entries. Both anomalies will require some additional preprocessing.

Ambiguous repayment status codes. The *PAY_X* columns are documented for -1 (paid on time) and $1-9$ (months of delay). In practice, values -2 and 0 also appear:

- -2 likely denotes months with no consumption
- 0 corresponds to revolving credit use

These undocumented codes are retained as-is, since they carry predictive information — clients with -2 or 0 are predominantly non-defaulters — and are handled correctly by the tree-based models adopted in this work.

Missing values. A full scan of the development set confirmed the absence of any missing values (NaN) across all 23 columns, so no imputation was necessary.

B. Exploratory Data Analysis

Pearson correlation analysis (Fig. 2) reveals that the repayment status columns are the dominant predictors: *PAY_0* achieves the highest correlation with the target ($|r| = 0.33$), followed by *PAY_2*–*PAY_6* in descending order. Demographic attributes (*sex*, *marriage*, *age*) and the credit limit show correlations below 0.15, suggesting limited individual discriminative power. The *BILL_AMT* columns are strongly inter-correlated (reflecting rolling monthly balances), motivating PCA as a candidate strategy for dimensionality reduction.

The distribution of the most recent repayment status (*PAY_0*) across both classes (Fig. 3) illustrates the separating power of payment delay: defaulters are concentrated in positive delay codes ($1-9$), while non-defaulters cluster at -1 (paid on time) or 0 (revolving credit). This pattern holds across all six *PAY* columns and directly motivates the delay-based engineered features described in Section II-A.

II. PROPOSED APPROACH

A. Preprocessing

Data cleaning. The column *education* contains undocumented values $\{0, 5, 6\}$ remapped to category 4 (“other”); *MARRIAGE* value 0 is remapped to 3 (“other”). No missing values were found.

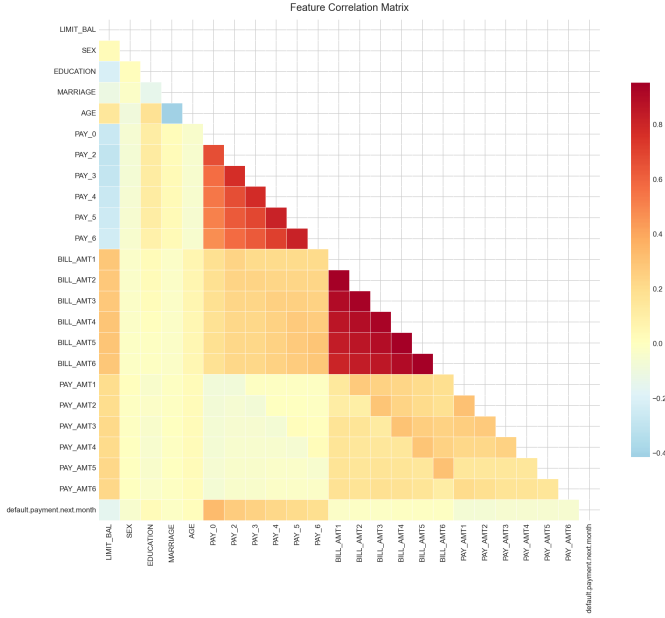


Fig. 2. Pearson correlation heatmap between all features and the target. Repayment status columns dominate; bill amounts are mutually correlated, motivating PCA as an alternative preprocessing strategy.

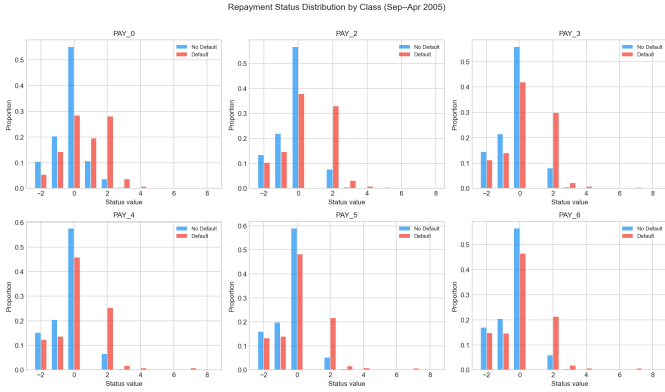


Fig. 3. Distribution of PAY_0 status values split by class. Positive delay codes are strongly associated with the default class.

Feature engineering (S3). Fifteen domain-specific features are derived and appended to the original 23 columns (Table I). The repayment status columns are the strongest individual predictors (max Pearson $|r| = 0.33$ with the target for PAY_0), so engineered features emphasize delay magnitude, trend, and payment coverage. This yields 38 features in total. Feature importance analysis (Fig. 4) confirms that engineered features such as *max_delay* and *avg_delay* rank alongside the raw PAY status columns as the most predictive signals.

One-hot encoding (S3+OHE). *sex*, *education*, and *marriage* are replaced by six binary dummy variables, bringing the total to 41 features. This has negligible impact on tree models (Δ Macro F1 < 0.001), as decision trees find optimal split thresholds regardless of ordinal encoding; it is nonetheless retained for representational correctness.

TABLE I
DOMAIN-SPECIFIC DERIVED FEATURES (S3 ENGINEERING).

Feature	Formula / Description
$UTIL_RATE$	$BILL_AMT1 / (-LIMIT_BAL + 1)$
$PAY_RATIO_i, i = 1..6$	$PAY_AMT_i / (-BILL_AMT_i + 1)$
MAX_DELAY	$\max(PAY_0, \dots, PAY_6)$
AVG_DELAY	$\text{mean}(PAY_0, \dots, PAY_6)$
$N_DELAYED$	# months with delay > 0
$DELAY_TREND$	$PAY_0 - PAY_6$
$BILL_TREND$	$BILL_AMT1 - BILL_AMT6$
$TOTAL_BILL/PAY$	$\sum BILL_AMT_i$ and $\sum PAY_AMT_i$
AVG_PAY_RATIO	$TOTAL_PAY / (-TOTAL_BILL + 1)$

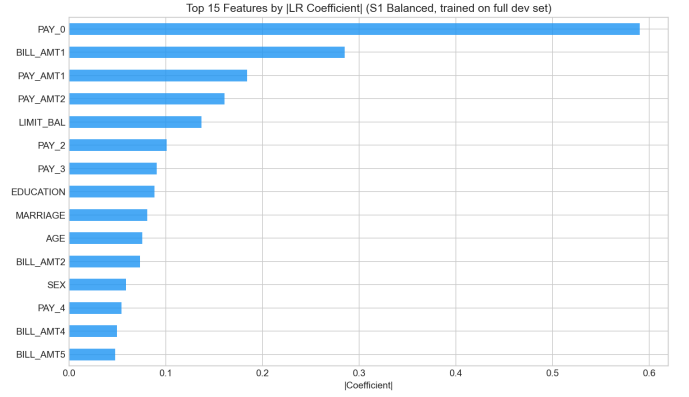


Fig. 4. Top feature importances from the tuned Random Forest (S3+OHE). Engineered delay features rank alongside raw PAY status columns, validating the S3 engineering strategy.

Class imbalance. *class_weight*='balanced' is applied to all relevant estimators, automatically up-weighting the minority class by the factor 3.52. SMOTE oversampling [2] was also evaluated but found inferior to feature engineering as a standalone strategy.

Six preprocessing pipelines are compared in Fig. 5 using a fixed Logistic Regression estimator. S3 (feature engineering + balanced weights) achieves the highest Macro F1 (0.677), outperforming PCA compression, SMOTE, and unbalanced training by a wide margin.

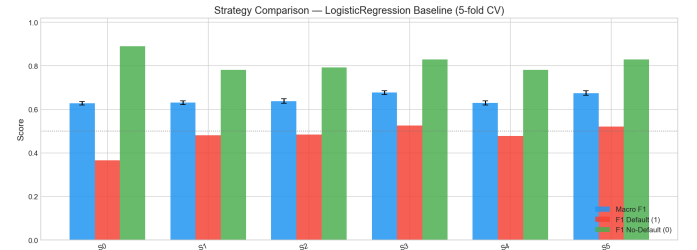


Fig. 5. Macro F1 (5-fold CV) across six preprocessing strategies with a fixed Logistic Regression. S3 (feature engineering + balanced weights) achieves the best result.

B. Model Selection

Six classifiers are benchmarked with the S3 preprocessing pipeline and five-fold stratified cross-validation (Table II).

Gradient Boosting and Random Forest [3] lead the ranking. Random Forest is $16\times$ faster (10 s vs. 166 s per CV) at only 0.005 Macro F1 below Gradient Boosting, making it the candidate of choice for Bayesian hyperparameter search. Naïve Bayes fails because of strong inter-feature correlations among *BILL_AMTX* columns, which violate its independence assumption.

TABLE II
MODEL COMPARISON WITH S3 PREPROCESSING (5-FOLD CV). RF IS SELECTED FOR TUNING ON THE BASIS OF ITS SPEED/PERFORMANCE TRADE-OFF.

Model	Macro F1	F1 Default	ROC-AUC
Gradient Boosting	0.684	0.477	0.779
Random Forest	0.679	0.467	0.770
Logistic Regression	0.677	0.525	0.758
KNN ($k=7$)	0.658	0.437	0.724
SVM (linear)	0.641	0.398	0.758
Naïve Bayes	0.397	0.398	0.737

The ROC curves (Fig. 6) confirm that Gradient Boosting and Random Forest dominate across all operating points, with Logistic Regression competitive at the high-recall end owing to better probability calibration. Naïve Bayes degrades sharply at low false-positive rates due to the correlated *BILL_AMTX* features as mentioned before.

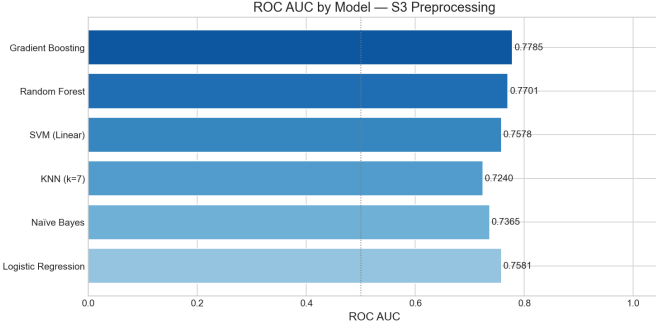


Fig. 6. ROC curves for all six classifiers with S3 preprocessing. Gradient Boosting (AUC=0.779) and Random Forest (AUC=0.770) lead across all operating points.

C. Hyperparameter Tuning

Bayesian optimization. Hyperparameters for both Random Forest and XGBoost [4] are optimized using Optuna’s Tree-structured Parzen Estimator (TPE) [5] with 5-fold stratified CV as the inner evaluation loop. The RF study runs 100 trials on S3+OHE features, warm-started from Phase 3 best parameters; XGBoost runs 50 trials on S3. Fig. 7 reports Macro F1 before and after tuning for both models across all three metrics: RF improves by +0.029 (0.679 $\rightarrow$ 0.707) and XGB by +0.020 (0.686 $\rightarrow$ 0.705). The tuned RF parameters are listed in Table III.

Probability threshold calibration. Because the default threshold $t = 0.5$ is not optimal for Macro F1 on imbalanced data a cross-validated sweep is performed over $t \in [0.25, 0.65]$

TABLE III
RF HYPERPARAMETERS BEFORE AND AFTER OPTUNA TUNING.

Parameter	Default (Phase 2)	Tuned (Phase 3/4)
$n_estimators$	300	650 / 350
max_depth	None	35 / 38
$min_samples_split$	2	9 / 9
$min_samples_leaf$	1	7 / 5
$max_features$	auto	sqrt
$class_weight$	None	balanced

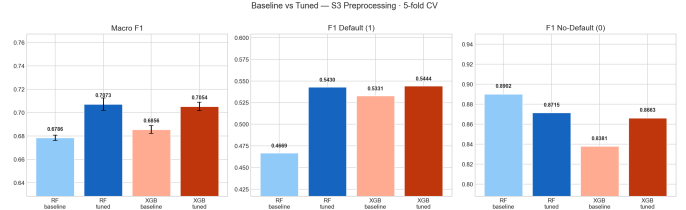


Fig. 7. Macro F1, F1-Default, and F1-No-Default for RF and XGBoost before (light) and after (dark) Bayesian tuning (5-fold CV). Both models improve substantially; RF achieves the best Macro F1.

in steps of 0.005: for each fold, the threshold maximizing Macro F1 on the held-out validation set is recorded; the mean across five folds is used as the final threshold. As shown in Fig. 8, the optimal threshold for the tuned RF lies near $t^* = 0.533$ (Phase 3 parameters, S3+OHE), yielding a further gain of +0.003 (0.707 $\rightarrow$ 0.711).

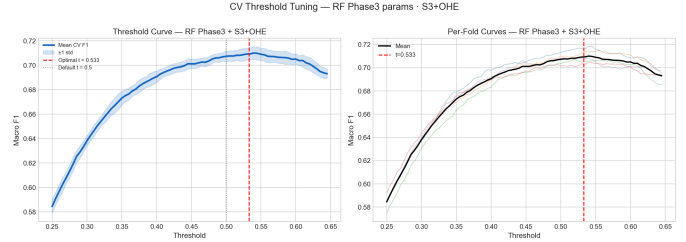


Fig. 8. CV threshold calibration curve for the RF model (Phase 3 params, S3+OHE). The shaded region shows ± 1 std across five folds. The Macro F1 peak is at $t^* = 0.533$, above the default 0.5.

III. RESULTS

Table IV reports Macro F1 at each experimental phase. The full pipeline (S3+OHE, tuned RF, $t = 0.533$) reaches a cross-validated Macro F1 of **0.711**, that represents a +0.083 absolute gain (+13.2% relative) over the unbalanced LR baseline. Each stage contributes incrementally: feature engineering provides the largest single step (+0.046), followed by Bayesian tuning (+0.029), and threshold calibration (+0.003).

The confusion matrix for the best model on the last CV fold (4,800 samples) is shown in Fig. 9. The model correctly identifies 57% of true defaulters (recall = 0.57) while maintaining high precision on non-defaulters (0.88). The remaining false negatives reflect the inherent difficulty of the task: many defaulters display payment patterns similar to non-defaulters up to six months before default.

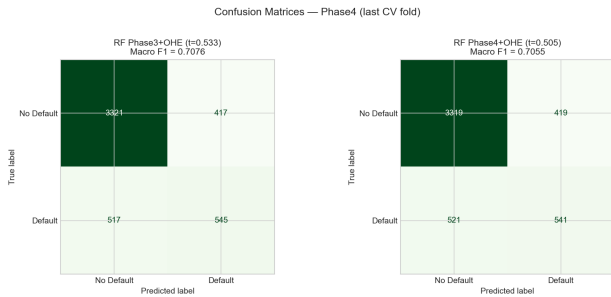


Fig. 9. Confusion matrices for the two best Phase 4 configurations on the last CV fold (4,800 samples). Both models reach Macro F1 ≈ 0.71 .

TABLE IV
MACRO F1 PROGRESSION ACROSS ALL EXPERIMENTAL PHASES (5-FOLD CV).

Phase	Configuration	Macro F1
Baseline	LR, no balancing	0.628
Balancing	LR + balanced weights	0.631
SMOTE	LR + SMOTE + balanced	0.638
Feature eng. (S3)	LR + S3 + balanced	0.677
Model selection	Random Forest, S3	0.679
Model selection	Gradient Boosting, S3	0.684
Tuning	XGB (S3, Optuna, $t=0.5$)	0.705
Tuning	RF (S3, Optuna, $t=0.5$)	0.707
OHE + t -calib.	RF (S3+OHE, $t=0.533$)	0.711
Submitted	RF (S3, $t=0.5$)	0.707

Despite the higher cross-validated score, the OHE + threshold-calibrated configuration generalised worse to the actual evaluation set than the simpler S3 + $t=0.5$ baseline. Because of this the final submission will be that one.

IV. DISCUSSION

Feature engineering is the dominant lever. The transition from scaled raw features to the S3 engineered set adds +0.046 Macro F1, more than six times the gain from switching classifiers (≤ 0.007) and comparable to the gain from Bayesian hyperparameter tuning (+0.029). This confirms a well-established pattern in tabular data modelling: domain knowledge encoded as hand-crafted features provides complementary signal that model selection cannot recover autonomously. The repayment status columns dominate the signal; derived features such as *max_delay*, *avg_delay*, and *pay_ratio_i* directly capture credit behavior that is not directly represented in the raw feature space.

OHE is negligible for tree-based models. Replacing the ordinal encoding of *sex*, *education*, and *marriage* with one-hot dummies changes Macro F1 by less than 0.001. Random Forests find optimal binary splits for integer-coded categoricals regardless of the numeric values assigned, so the ordinal representation is equally effective. The experiment nevertheless validates that no information is lost in the encoding.

Threshold calibration. Shifting the classification threshold from 0.5 to 0.533 yields +0.003 Macro F1 on cross-validation,

indicating that the model slightly over-predicted the *Default* class at the default cutoff. However, this CV gain did not translate to the actual evaluation set: the calibrated threshold underperformed $t = 0.5$, suggesting that the optimization overfit to the development distribution (see the following paragraph). The default threshold $t = 0.5$ was therefore retained for the final submission.

CV score does not always predict submission performance. The OHE + calibrated-threshold configuration achieved the highest cross-validated Macro F1 (0.711) yet underperformed the simpler S3 + $t=0.5$ model on the actual evaluation set. Two likely causes:

- (i) the threshold $t^* = 0.533$ was optimized on the same 24 000-sample development set used for model selection, introducing a systematic optimistic bias that compromised the nested CV design;
- (ii) one-hot encoding the three low-signal categorical variables (*sex*, *education*, *marriage*) adds six extra splits that a deep Random Forest can over-exploit, slightly inflating CV performance on this particular sample.

The result highlights the importance of validating complexity gains on a truly held-out partition before treating CV improvements as reliable.

Limitations and future work. The best model still achieves $F1 \approx 0.54$ on the minority (*Default*) class, reflecting the inherent difficulty of the task. Promising avenues for further improvement include: ensemble stacking (RF + XGB + LR meta-learner), LightGBM as a faster boosting alternative, *RobustScaler* for right-skewed bill and payment columns, and additional interaction features (e.g., polynomial *PAY₀* terms or ratios across months).

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